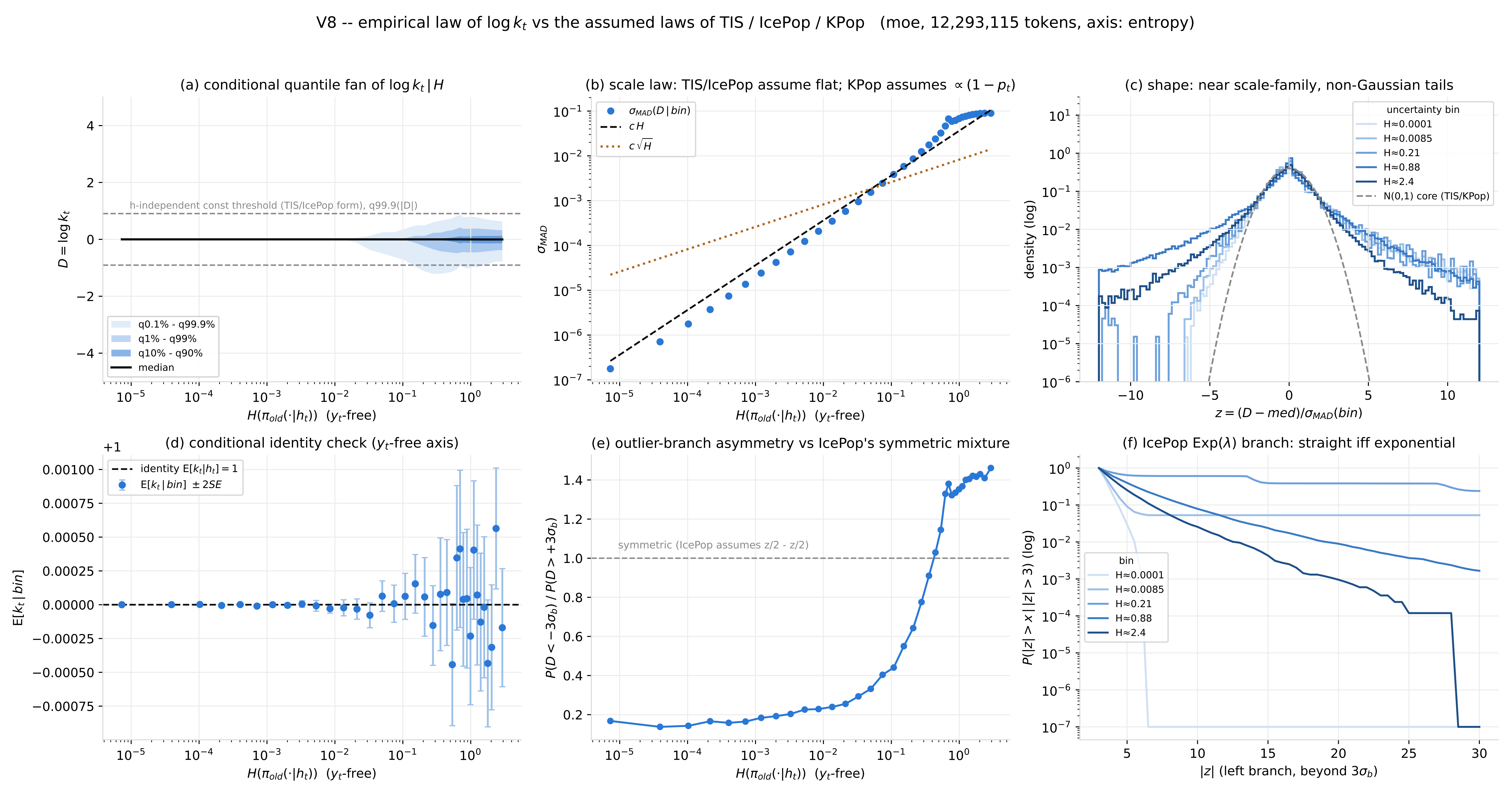
V8 -- empirical law of log k_t vs the assumed laws of TIS / IcePop / KPop (moe, 12,293,115 tokens, axis: entropy)
(a) conditional quantile fan of $\log k_t \mid H$
h-independent const threshold (TIS/IcePop form), q99.9(|D|)
$D = \log k_t$
q0.1% - q99.9%
q1% - q99%
q10% - q90%
median
$H(\pi_{old}(\cdot|h_t))$ (y_t-free)
(b) scale law: TIS/IcePop assume flat; KPop assumes $\propto (1 - p_t)$
$\sigma_{MAD}(D \mid bin)$
$c\,H$
$c\sqrt{H}$
σ_{MAD}
$H(\pi_{old}(\cdot|h_t))$ (y_t-free)
(c) shape: near scale-family, non-Gaussian tails
uncertainty bin
H≈0.0001
H≈0.0085
H≈0.21
H≈0.88
H≈2.4
N(0,1) core (TIS/KPop)
density (log)
$z = (D - med)/\sigma_{MAD}(bin)$
(d) conditional identity check (y_t-free axis)
identity $E[k_t|h_t] = 1$
$E[k_t \mid bin] \pm 2SE$
$E[k_t \mid bin]$
$H(\pi_{old}(\cdot|h_t))$ (y_t-free)
(e) outlier-branch asymmetry vs IcePop's symmetric mixture
symmetric (IcePop assumes z/2 - z/2)
$P(D < -3\sigma_b) / P(D > +3\sigma_b)$
$H(\pi_{old}(\cdot|h_t))$ (y_t-free)
(f) IcePop Exp(λ) branch: straight iff exponential
bin
H≈0.0001
H≈0.0085
H≈0.21
H≈0.88
H≈2.4
$P(|z| > x \mid |z| > 3)$ (log)
$|z|$ (left branch, beyond $3\sigma_b$)